

COURSE TITLE : KNITTING & NON-WOVEN FABRICS
COURSE CODE : 6062
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 75
CREDIT : 5

TIME SCHEDULE

Module	Topics	Period
1	Manufacture of Non - Wovens	17
2	Knitted Structure	20
3	Warp & Weft Knitting	20
4	General study of Technical Textile	18
Total		75

COURSE OUTCOME

Module	Go	Student will be able to
1	1	Understand fibre preparation & web formation in non-woven industry
	2	Know manufacturing methods & uses of non-woven fabrics
2	1	Understand knitting & terms related to knitting
	2	Know different knitting elements
	3	Understand about various knitting structures
3	1	Understand different weft knitting machines and their features
	2	Know Rachel and Tricot warp knitting machines & the features
4	1	Understand features of Industrial Textiles
	2	Know Geo-Textiles, agricultural & medical textiles

SPECIFIC OUTCOME

MODULE II MANUFACTURE OF NON - WOVENS

1.1.0 Understand fibre preparation & web formation in non-woven industry

- 1.1.1 State the definition and classification of non-woven
- 1.1.2 Compare woven and non-woven fabrics
- 1.1.3 List important fiber properties used for the manufacturing of non-woven
- 1.1.4 Identify various stages involved in non woven fabric production
- 1.1.5 Describe the fibre preparation techniques
- 1.1.6 Explain various web forming methods

1.2.0 Know manufacturing methods & uses of non-woven fabrics

- 1.2.1 Explain the production of needle punched fabrics
- 1.2.2 Explain the Production of thermal bonded fabrics
- 1.2.3 Illustrate chemical bonding techniques
- 1.2.4 List the properties and application of non-woven fabrics

MODULE II KNITTED STRUCTURES

2.1.0 Understand knitting & terms related to knitting

- 2.1.1 State the knitting process
- 2.1.2 List types of knitting
- 2.1.3 Compare weaving and knitting
- 2.1.4 List and define the terms related to knitting

2.2.0 Know different knitting elements

- 2.2.1 List various knitting elements
- 2.2.2 Describe the knitting elements
- 2.2.3 Explain various knitting needles
- 2.2.4 Illustrate Knitting actions of various needles

2.3.0 Understand about various knitting structures

- 2.3.1 Define Knitted structure
- 2.3.2 List & illustrate various knitted structures
- 2.3.3 Describe the properties of plain Rib and Interlock fabrics

MODULE III WARP & WEFT KNITTING

3.1.0 Understand different weft knitting machines and their features

- 3.1.1 List types & features of weft knitting
- 3.1.2 Explain the application of weft knitted fabrics
- 3.1.3 Describe the passages of yarn through Single Jersey circular weft knitting machine
- 3.1.4 Explain the passage of yarn through V - bed flat knitting machine
- 3.1.5 List the properties of weft knitted fabrics

3.2.0 Know Rachel and Tricot warp knitting machines & the features.

- 3.2.1 Define warp knitting and its features
- 3.2.2 Explain the application of warp knitted fabrics
- 3.2.3 Describe knitting action of Rachel warp knitting machine
- 3.2.4 Explain knitting action of Tricot warp knitting machine
- 3.2.5 List the properties of warp knitted fabrics.

MODULE IV GENERAL STUDY OF TECHNICAL TEXTILE

4.1.0 Understand features of Industrial Textiles

- 4.1.1 Classify Non-apparel textiles
- 4.1.2 Define Industrial Textiles
- 4.1.3 Explain the scope and application of Technical Textiles
- 4.1.4 State the features of industrial textiles
- 4.1.5 Describe automotive textiles, Filter fabrics, Conveyor fabrics, Tarpaulins , protective fabrics

4.2.0 Know Geo-Textiles, agricultural & medical textiles

- 4.2.1 Define geo-textiles
- 4.2.2 State the features of Geo textiles
- 4.2.3 Describe the functions of Geo textiles
- 4.2.4 Illustrate the applications of Geo textiles
- 4.2.5 Describe the agricultural textiles and its applications
- 4.2.6 Define medical textiles
- 4.2.7 Explain the types and application of medical textiles

CONTENT DETAILS

MODULE I - MANUFACTURE OF NON-WOVENS

Definition of non-wovens- Classification of Non-wovens-Stages involved in the production of Non – woven fabrics- Comparison between woven and non-woven fabrics- Basic requirements of fibers used for the manufacture of non-wovens -Methods of web-formation – card web - rando feeder –spun bond - Different methods of web laying – Reinforcement--Various methods of Fiber bonding techniques - Mechanical bonding-Chemical bonding-Thermal bonding-Importance of needle punching - Study of needling loom-stitch bonding with threads & without threads - Study of various methods of chemical bonding-Study of various methods of Thermal bonding- the important properties of non-woven fabrics-Discuss the limitations of non-woven fabrics- Application of Non-woven fabrics

MODULE II - KNITTED STRUCTURE

Definition of Knitting-Classification of knitting-Compare woven and knitted structures-Compare the process of Knitting and weaving-Define the term-Courses-Wales-Gauge-Face loop-Back loop-open loop – closed loop-Loop length-Texture-Single Jersey-Double Jersey - Study of various knitting elements-Latch needle-Bearded Needle-Compound Needle-cylinder-dial - Sinker-Cams-Study of knitting action of various types of needles- Define knitted structures-Plain knit-rib knit-Interlock - Study the properties of Plain, Rib, Interlock- Define Knit stitch-Miss stitch-Tuck stitch.

MODULE III - WARP & WEFT KNITTING

Define weft knitting-Application of weft knitted fabrics - Passage of Yarn through single Jersey weft knitting machine - parts of circular knitting m/c - Passage of yarn through V-bed flat knitting machine- Warp knitting structure - Features of warp knitting- Application of warp knitted fabrics - Different types of knitting elements-Study of knitting actions of Rachel Warp knitting machine-Study the knitting actions of Tricot warp Knitting machine.

MODULE IV - GENERAL STUDY OF TECHNIAL TEXTILES

Classify Non-apparel textiles-Identify the application of Technical Textiles - Study the scope and application of Technical Textiles-Brief study of Industrial Textiles – automotive textiles-tyre fabrics, filter fabrics, conveyor fabrics, tarpaulins and coverings - protective fabrics-bullet proof fabrics, fire proof fabrics, agriculture fabrics, parachute fabrics-study of Geo Textiles-Application of Geo Textiles - Medical Textiles – Hygiene fabrics .

REFERENCE BOOKS

- | | | |
|-----------------------|----------------------------------|----------------------------|
| 1. P.W.Moncrieff | Man-made Fibres, | Newens Buttesworth London |
| 2. V.A.Shenoi | Textile Fibres Vol-I | Sevak publication |
| 3 . David.J.Spencer | Knitting Technology | Academic publication |
| 4. J.A.Smirfitt | An Introduction to weft knitting | Marrow publication England |
| 5. Thomson | An Introduction to warp knitting | Marrow publication England |
| 6. Prof. N.N.Banerjee | Knitting Technology | - |
| 7. ATIRA | Technical Textiles | - |
| 8. Ajagoankar | Knitting technology | - |